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Lecture 3: Malware and Trojans

PART A: Malware

1. Introduction to Malware

Malware (Malicious Software) is any software intentionally designed to **disrupt, damage, spy on, or gain unauthorized access** to computer systems, networks, or data.

Malware violates one or more goals of the **CIA Triad**:

- Confidentiality (data theft)
- Integrity (data modification)
- Availability (system disruption)

2. Malware Infection

Malware infection refers to the process by which malware enters, spreads, and executes on a target system.

Common Infection Methods

- Email attachments and phishing links
- Malicious websites (drive-by downloads)
- Infected USB drives
- Pirated or cracked software
- Software vulnerabilities

3. Components of Malware

A typical malware consists of the following components:

3.1 Infection Mechanism

Determines **how malware enters the system**

Example: email attachment, exploit kit



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3.2 Payload

The **malicious action** performed by malware

Examples:

- Data theft
- File deletion
- Encryption (ransomware)

3.3 Command and Control (C2)

Enables communication between the malware and the attacker's server.

3.4 Persistence Mechanism

Ensures malware **remains active after reboot**.

3.5 Propagation Mechanism

Allows malware to **spread to other systems**.

4. Types of Malware

4.1 Virus

- Attaches itself to a host file
- Requires user action to execute
- Spreads when infected file is shared

4.2 Worm

- Self-replicating malware
- Spreads without user interaction
- Consumes network bandwidth

4.3 Trojan Horse

- Disguised as legitimate software
- Does not self-replicate
- Creates backdoors



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4.4 Ransomware

- Encrypts victim's files
- Demands ransom payment

4.5 Spyware

- Collects sensitive information
- Monitors user activity

4.6 Adware

- Displays unwanted advertisements
- May track browsing behavior

4.7 Rootkit

- Hides malware presence
- Gains privileged access

4.8 Bot / Botnet

- Turns infected system into a zombie
- Controlled remotely by attacker

PART B: Trojans

5. What is a Trojan?

A **Trojan Horse** is a type of malware that **appears legitimate** but performs **hidden malicious actions** once executed.

Key characteristics:

- Disguised as trusted software
- Requires user execution
- Opens backdoors
- Enables remote control



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6. Indications of Trojan Attack

Common signs of Trojan infection include:

- Slow system performance
- Unknown programs running
- Unexpected pop-up windows
- Disabled antivirus software
- Unusual network traffic
- Unauthorized access attempts

7. How Hackers Use Trojans

Hackers use Trojans to:

- Steal login credentials
- Monitor keystrokes (keyloggers)
- Install additional malware
- Control victim's system remotely
- Participate in botnet attacks

8. Types of Trojans

8.1 Backdoor Trojans: Provide unauthorized remote access

8.2 Remote Access Trojans (RATs): Full control over victim's system

8.3 Banking Trojans: Steal financial credentials

8.4 Downloader Trojans: Download additional malware

8.5 Spy Trojans: Monitor user activities

8.6 Rootkit Trojans: Hide malicious processes

8.7 DDoS Trojans: Launch distributed denial-of-service attacks